SMART EMERGENCY PATIENT COORDINATION SYSTEM

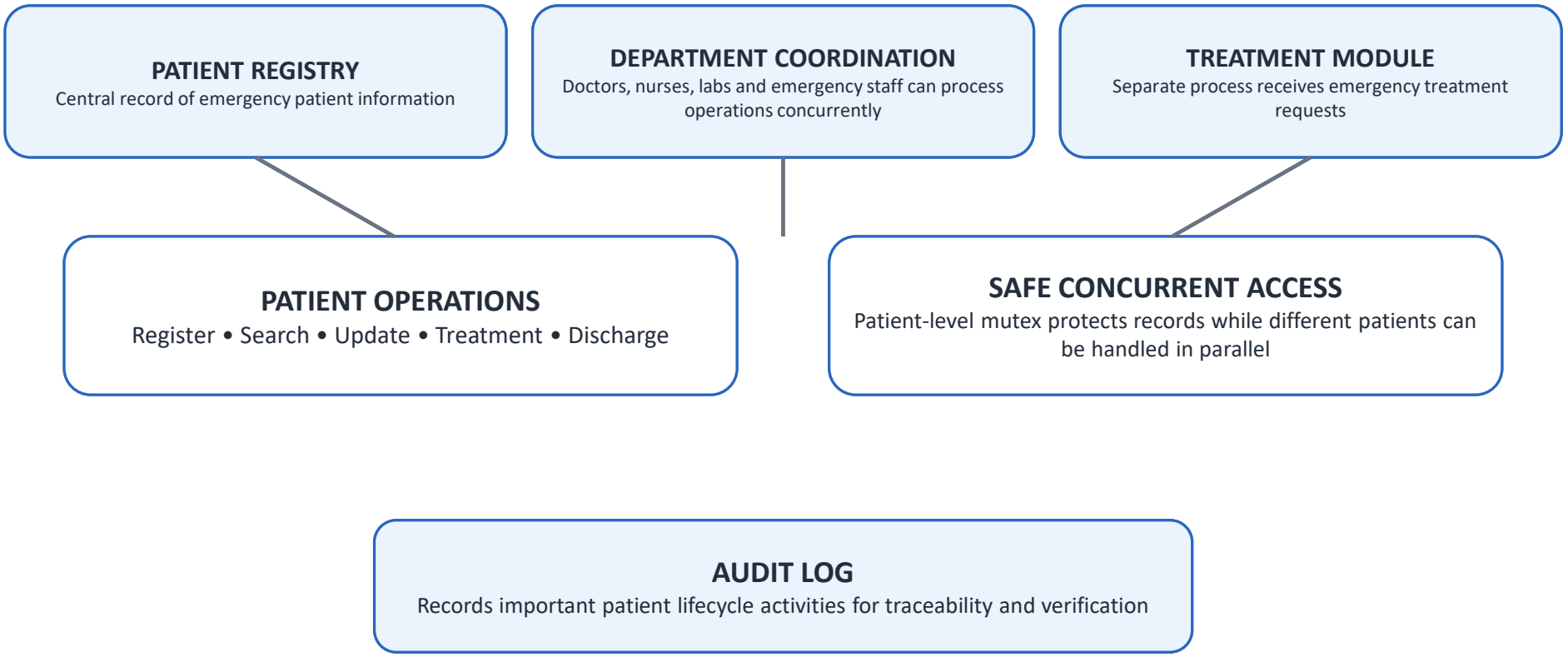
System Architecture & Emergency Coordination Prototype

ARCHITECTURE OVERVIEW

System Core • Problem • Architecture • Critical Conditions • Solution • End-to-End Flow

1. System Core

What the system manages and coordinates



2. Problem Layer

Challenges the architecture is designed to address

1. CONCURRENT ACCESS
Multiple departments may access the same patient record at the same time.

2. COARSE LOCKING
A single global lock can block unrelated patient operations and reduce scalability.

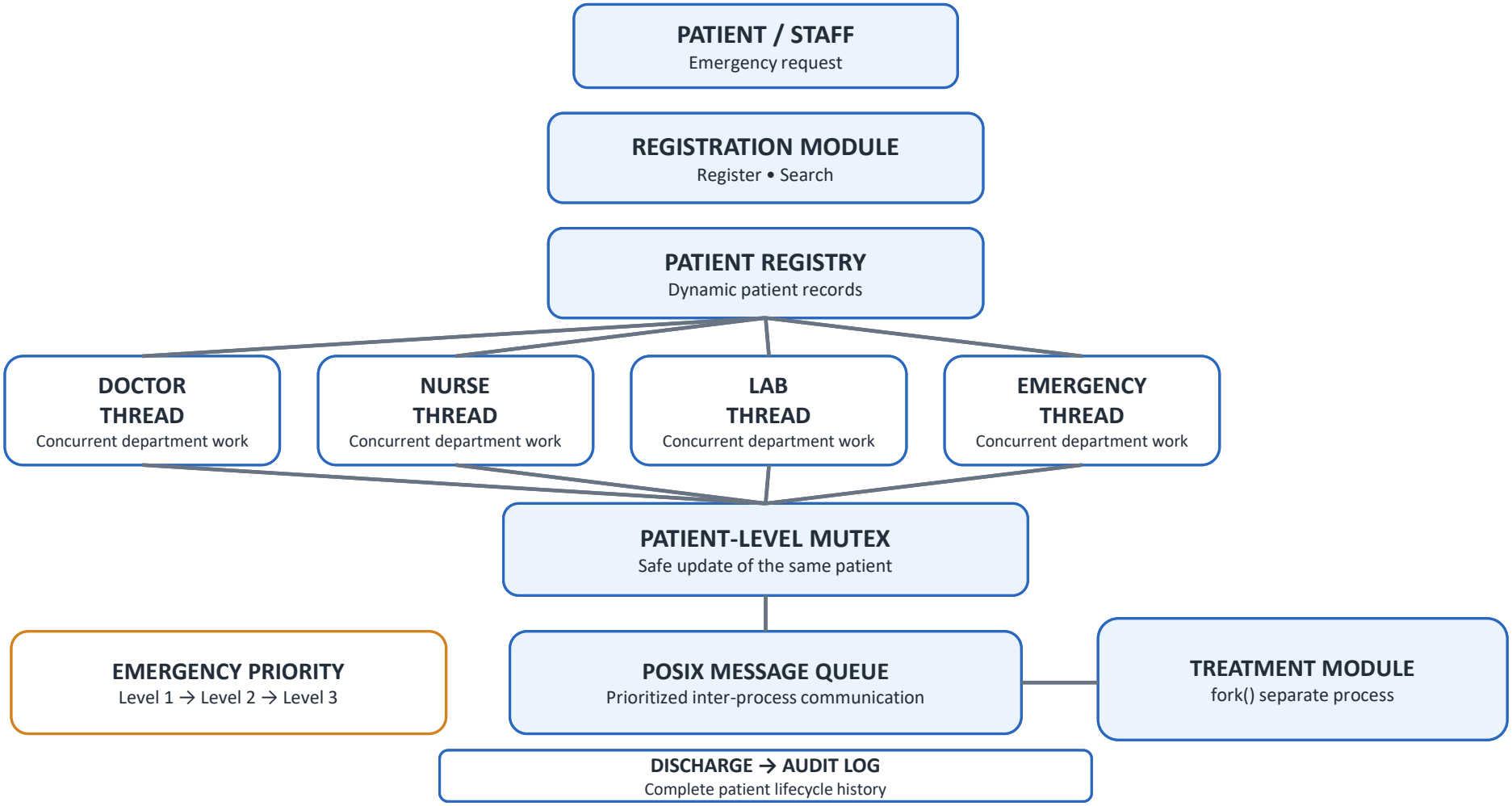
3. UNSYNCHRONIZED UPDATES
Concurrent read–modify–write operations may overwrite another department's update.

4. CRITICAL CASES
Emergency requests need priority handling so urgent cases are processed first.

5. TRACEABILITY
Registration, updates and discharge need an activity history for accountability and verification.

IMPACT
Waiting • Conflicting updates • Inconsistent records • Delayed emergency handling • Limited traceability

3. Complete System Architecture Diagram



4. Critical Conditions

Priority-based emergency treatment handling

LEVEL 1
CRITICAL / LIFE-THREATENING
Highest IPC priority
Immediate treatment dispatch

LEVEL 2
SERIOUS EMERGENCY
High priority
Process after Level 1

LEVEL 3
LESS CRITICAL
Normal processing
Lower priority

PRIORITY DECISION

Emergency condition detected → assign level → send prioritized request → Treatment Module processes urgent cases first

5. Solution Layer

Design choices that address the identified problems

FINE-GRAINED SYNCHRONIZATION
One independent mutex for each patient record.

PARALLEL PROCESSING
Different patient records can be processed concurrently.

SAFE SERIALIZATION
Updates to the same patient are serialized safely.

SEPARATE TREATMENT PROCESS
fork() creates a separate Treatment Module.

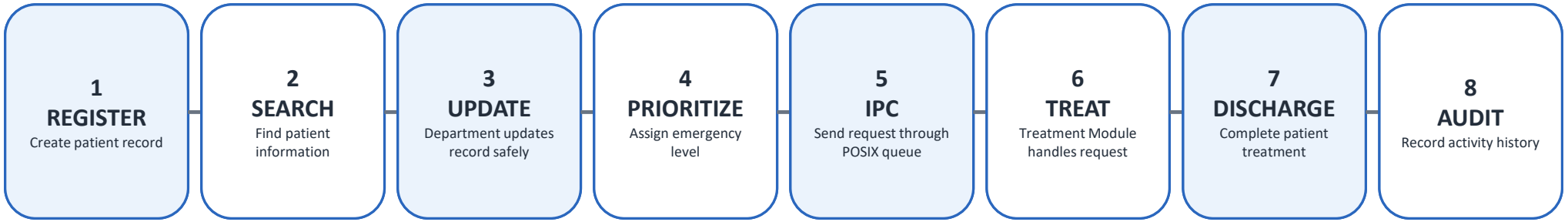
PRIORITY IPC
POSIX message queues carry prioritized emergency requests.

AUDITABILITY
File-based history records important patient lifecycle events.

RESULT
Safe + consistent updates • Faster parallel work • Priority treatment • Traceable operations

6. End-to-End System Flow

Complete emergency patient lifecycle



FINAL SYSTEM OUTCOME

A coordinated emergency workflow that protects patient records, prioritizes critical requests, supports concurrent department work, and maintains traceability.